

2.1 Sequential Models and RNNs

A shared cell, then an RNN-LM: next-token softmax from the hidden state.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

An MLP sees a vector and forgets. Language is a **sequence**: the meaning of *not* depends on what came before. A recurrent net keeps a **state** and updates it at every token, with the **same weights**.

This week is the historical path to transformers. You still need it: LSTMs show up in speech and time series, and the failure mode (a chain of Jacobians) is why Week 3 exists.

First time this method appears. An **RNN** is a net that shares one cell across time.

What. Hidden state h_t is a function of h_{t-1} and the new token. An RNN-LM predicts the next token from h_t . **Why.** A fixed window (n-gram / bag) cannot count “the verb agrees with the noun 40 tokens back” without growing width. **Architecture.** Shared cell $h_t = \tanh(W_{hh}h_{t-1} + W_{xh}x_t + b)$, then softmax on h_t . Unroll in time for training (BPTT). **How.** Read left to right. The same W at every step. Lab 2 copies the first bit at the end. **Formula.** $p(w_t | w_{<t}) = \text{softmax}(W_o h_t)_w$. **Tradeoffs.** + Variable length, parameter sharing. – Sequential (hard to parallelize), vanishing/exploding gradients (next note).

1. Why a fixed window is not enough

A language model predicts the next token. Week 1 already wrote

$$P(w_1, \dots, w_T) = \prod_{t=1}^T P(w_t | w_{1:t-1}).$$

An **n-gram** approximates the history by the last $n - 1$ words and **counts**. Raise n and the table is sparse; most strings never occurred. A **fixed-window MLP** concatenates the last k embeddings and is dense, but k is still a hard cutoff and each position in the window has its own weights.

You need a map that (i) accepts **any length**, (ii) **shares** weights across time, and (iii) can, in principle, use the beginning of the sentence. That map is a recurrent cell.

Do not build an n-gram this week. The counting story is only here so the cell has a job.

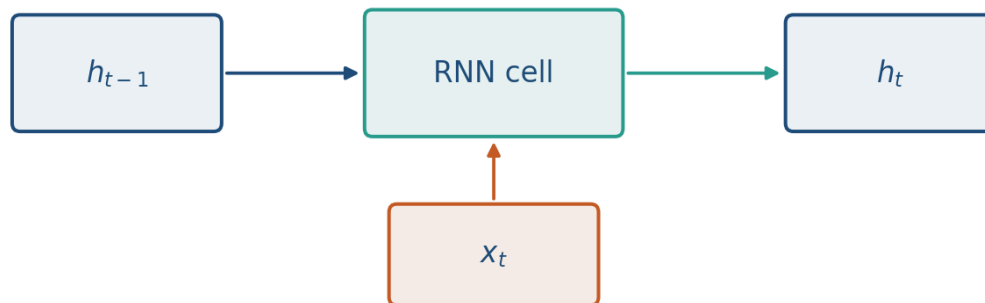
2. The cell

At step t the cell takes

- the new token embedding x_t , and
- the previous hidden state h_{t-1} .

It emits h_t , which is supposed to hold **both**. That vector is the memory.

Same weights at every step: input plus previous state



The vanilla (Elman) recurrence is

$$h_t = \tanh(W_h h_{t-1} + W_x x_t + b) .$$

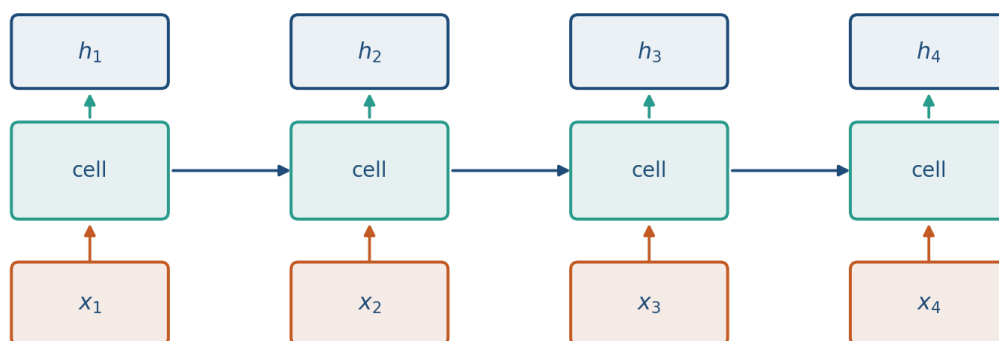
The same W_h, W_x, b are used at every step. That is weight sharing along time, the sequence analogue of a CNN's shared filter.

Between two independent sequences (two reviews in a batch) you **reset** $h_0 = 0$. The loop lives *inside* one sequence. If you do not reset, review 2 starts with leftover sentiment from review 1. That is leakage, not transfer learning.

3. Unrolled in time

Write the loop as a chain. Four tokens, one set of weights:

Unrolled in time. The chain is why long-range gradients struggle.



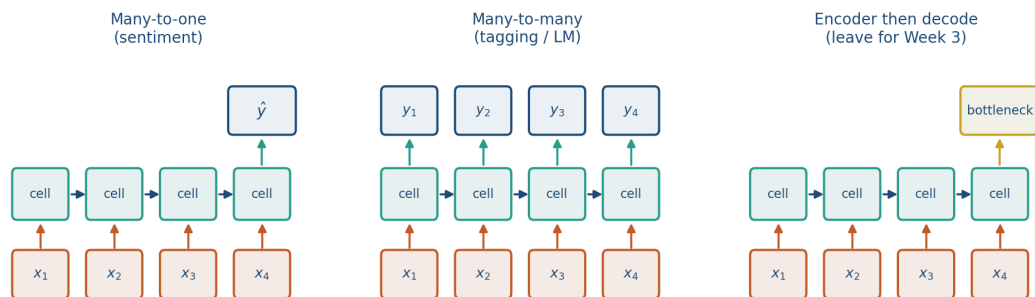
A CNN looks at a **fixed** neighborhood, in parallel. An RNN looks at **everything so far**, in order, and cannot fully parallelize across time. That is the accuracy / speed tradeoff.

Unrolling is how you backprop (note 2.2). The picture is one graph with **copies** of W_h . There are not four different matrices named $W_h^{(1)}, W_h^{(2)}, W_h^{(3)}, W_h^{(4)}$.

If the first word is the subject of a verb twenty tokens later, h_t must still contain that subject. The cell is a lossy compressor: every step overwrites. Week **2.2** is what happens to the gradient on that long path. Week **2.3** adds gates so the cell can choose what to keep.

4. What you can wire

The same cell, different wiring:



- **Many-to-one** (sentiment): read $x_1, \dots, x_T$, emit from the last h_T . Lab 2 is this shape: copy the first bit at the end.
 - **Many-to-many, aligned** (tagging, language modeling): emit at every t .
 - **Encoder then decode**: compress the source into one vector, then generate. That bottleneck is why Week 3 exists. Do not train seq2seq this week.
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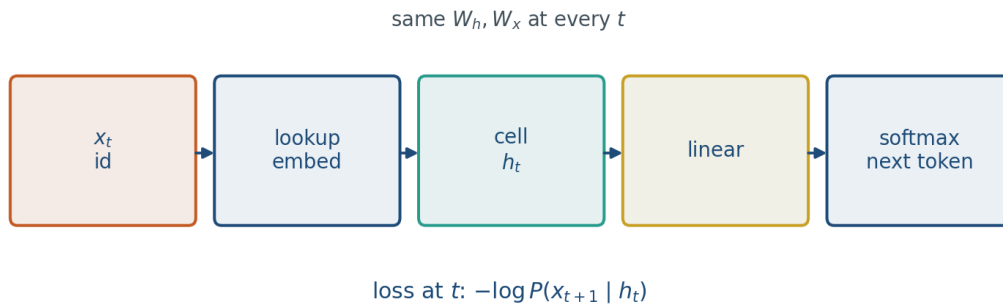
5. An RNN language model

Lookup the token id, run the cell, then a linear map plus **softmax** over the vocabulary. The loss at step t is the same next-token NLL as Week 1:

$$L_t = -\log P(x_{t+1} | h_t).$$

Average over tokens in the **sentence** (or a small batch of sentences). That is teacher forcing: the true previous token is the next input, not the model's sample.

An RNN language model is next-token softmax from the hidden state



Perplexity is $\exp(\text{mean NLL})$ in nats, or $2^{\text{mean NLL}}$ if you used log base 2. Lower is better. You will not chase a Wikipedia perplexity number this week. You will chase whether Lab 2's first bit is still in h_T .

Generation (optional picture): sample $\hat{x}_{t+1}$ from the softmax, feed it back as x_{t+1} . Training does not do that.

6. Teaching this note

~**18 minutes**. Why a window is not enough, then the cell, then unroll three tokens with **shared** W_h , then the 2-D numeric table, then the RNN-LM diagram so next-token NLL has a hidden state. Play **0:00–12:00** of StatQuest RNNs (the loop and the unroll). Save vanishing gradients for note 2.2 even if the video teases them.

7. Worked example

Let $d = 2$, $b = 0$, $h_0 = 0$,

$$W_h = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}, \quad W_x = I, \quad x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Step 1: $W_h h_0 + W_x x_1 = x_1$, so $h_1 = \tanh(x_1) \approx \begin{bmatrix} 0.76 \\ 0 \end{bmatrix}$ because $\tanh(1) \approx 0.76$.

Step 2: $W_h h_1 + W_x x_2 \approx \begin{bmatrix} 0.38 \\ 1 \end{bmatrix}$, then $h_2 = \tanh(\cdot) \approx \begin{bmatrix} 0.36 \\ 0.76 \end{bmatrix}$.

$W_h = 0.5I$, $W_x = I$, $h_0 = 0$. Token 1 shrinks in h_2 .

step	x_t	$W_h h_{t-1} + W_x x_t$	$h_t = \tanh(\cdot)$
$t = 1$	[1, 0]	[1, 0]	[0.76, 0]
$t = 2$	[0, 1]	[0.38, 1]	[0.36, 0.76]

The same W_h was used twice. Token 1 still has a **shrunk** footprint in h_2 ($0.76 \rightarrow 0.36$). That shrinkage is the teaser for vanishing gradients.

8. Where students get stuck

- Drawing a new W_h at each time step (no sharing).
 - Feeding the whole sentence as one concatenated vector “to make it an MLP.”
 - Forgetting to reset h between batch items.
 - Thinking an RNN-LM samples during training. It uses the true previous token.
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9. Video

Watch [StatQuest: Recurrent Neural Networks, Clearly Explained](#).

Pause on the first unroll (same weights copied) and on the moment the hidden state is described as memory. Skip any long software demo; the board math is the point.

10. Practice

1. Unroll three tokens on paper with $d = 2$, $h_0 = 0$. Which matrices are reused?
2. Why reset the state between two unrelated documents in a batch?
3. Name one thing an RNN can mix that a kernel of size 5 cannot, in a single layer.
4. Using the worked example's W_h , W_x and $x_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, write the pre-activation $W_h h_2 + W_x x_3$ with $h_2 \approx \begin{bmatrix} 0.36 \\ 0.76 \end{bmatrix}$. Do not bother with $\tanh$ if you are short on time; the linear mix is the lesson.
5. If every W_h were replaced by $0.1I$, and $x_t = 0$ after $t = 1$, approximately how does $\|h_t\|$ scale with t ? (Geometric sequence.)

6. In one sentence: what is the loss at step t of an RNN language model? What is teacher forcing?
7. Many-to-one or many-to-many: which wiring is Lab 2, and which wiring is next-token language modeling?